MYRRHA WCS Engineering Handover — SUMMARY (v0.4)

Project Overview

Engineering handover bundle for the **Warm Compressor Station (WCS)** of the MYRRHA cryogenic plant (QPLANT). Compares the two pre-studied warm-compressor concepts — Kaeser **FSD 575 SFC** (ALaT) vs **FSD 475 SFC** (LKT) — against a **350 g/s @ 14 barG** baseline, sizes utilities, and consolidates the operational envelope under a Greek-letter notation (β / γ / α).

Location: `/home/ubuntu/myrrha_handover/` · **Entry point:** `index.html`

Deliverables

HTML views (13)

#	File	Purpose
1	`index.html`	Hub / navigation
2	`slides.html`	Interactive deck
3	`slide-editor.html`	Dual-view live editor
4	`reports.html`	Rendered design arguments B1–B7
5	`tables.html`	Specs · CSV / JSON export
6	`graphs.html`	Chart.js plots
7	`status.html`	Session TODOs · changelog
8	`truth.html`	Immutable design decisions
9	`utilities.html`	Math · sample calculations
10	`pvps.html`	2 × 5 LP array · γ = 50 g/s · N+1
11	`hx-sizing.html`	RCW oil cooler · COMBI · Plotly
12	`thermo-analysis.html`	Energy / Sankey / T-s / exergy
13	`operations.html`	β/γ/α operating windows · 3D surface
14	`handover.html`	Agent BIOS · session protocol

Master sources (single source of truth)

- `master/data.json` — all numeric data
- `master/slides.md` — slide content
- `master/truth.md` — design decisions
- `master/config.yaml` — themes / fonts / layout

Pipeline & docs

- `render_all.py` , `live-render.js`
- `docs/design-guide.md` , `docs/editing-guide.md` , `docs/slide-notes-guide.md`

Key Findings & Recommendations

1. **Baseline:** 350 g/s @ 14 barG holds as the utility-sizing point (β_max).
2. **Vendor selection:** LKT (3 × FSD 475) preferred over ALaT (2 × FSD 575) for finer turndown granularity and N+1 economy.
3. **PVPS:** 2 × 5 LP-unit array (γ_nominal = 50 g/s) provides α = 0.4–1.0 turndown with one-unit redundancy.
4. **RCW:** ΔT 25 °C (30 → 55 °C) on oil pre-cool; HX N+1 strategy pending approval.
5. **COMBI concept:** Fixed 64 Hz + VFD unit promising for Config24 — needs Kaeser confirmation.
6. **Exergy efficiency:** ~44 % (current target); review for v0.5.

Operational Notation (recap)

- **β** = WCS.HP flow [g/s], β_max = 350
- **γ** = PVPS flow [g/s], γ_nominal = 50
- **α** = γ / γ_max, α ∈ [0.4, 1.0]
- **Σ** = β + γ (total WCS)
- Design points: **Config30** (β=330, Σ=380) · **Config24** (β=304, Σ=354)

Next Steps for Engineering Team

1. **Critical (blocking):** Coldbox QRB interface · COMBI vendor confirmation.
2. **High:** PVPS datasheets · turndown scenarios C30(00,10) / C24(00,10) · RCW HX N+1.
3. **Medium:** Discharge temperatures · noise compliance · waste-heat recovery.
4. **Low:** HVAC allocation · exergy optimization.

Handover Instructions

- **Boot point:** open `handover.html` for agent BIOS and session protocol.
- **Edit content:** modify files in `master/` ; run `render_all.py` to regenerate views.
- **Versioning:** v0.4 = 85 % complete; v0.5 will integrate vendor data + QRB coldbox.

Contact

Direct continuation in this conversation, or load `handover.html` § “Next-session boot prompt” into a new agent session.